Jamie Seoh

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Education

University of Waterloo Sep 2025 – May 2030
Bachelor of Software Engineering *3.95/4.00 (91.2%) GPA*

- Top 7.6% in Euclid Math Contest, 2 x Hackathon Winner, $8K in Scholarships

Technical Skills

Hardware: C/C++, SystemVerilog, Python, Bash, RISC-V, MIPS, Arduino, Teensy, Scala
Software: LLVM, PyTorch, Numpy, TypeScript/JavaScript, HTML/CSS, SQL, React, FastAPI
Tools: Linux/Unix, Git/GitHub, Docker, Altera Quartus Prime, Fusion 360, Jira/Confluence, Figma

Experience

Cerebras Systems May 2026 – Aug 2026
AI Software Engineer Intern *Toronto, ON*

- Engineered a **fault-injection** tool with **C++** and **assembly** spanning runtime controls and on-device kernels to simulate crashes, stalls, and slowdowns, reducing average customer-failure triage time by **2 days**.
- Refined fault detection to separate likely faulty devices from affected neighbors, **reducing major reports** flagged for on-call escalation from **11 to 1** in one production-stall replay.
- Built a **breakpoint** managing **concurrent requests** and exposing counters to debug a fatal KV-cache refill bug.
- Added deterministic **GTest** & **pytest** regression tests to catch unintended worker pauses and verify delay timing.
- Developed a worktree and agent management dashboard to streamline parallel development for fellow interns.

University of Waterloo Sep 2026 – Present
Undergraduate Research Assistant *Waterloo, ON*

- Researching **Rust-to-LLVM** IR translation and verification semantics under Prof. Arie Gurfinkel to extend Rust support for SeaHorn, a tool for automated program verification.
- Wrote verification harnesses in Rust for **pointer aliasing** and memory updates; repaired unreachable and compiler-eliminated assertions, validated through vacuity and mutation checks.

Projects

Self-Playing MIDI Keyboard | *C++, Arduino, Teensy 4.0*

- Designed a 25-key MIDI keyboard with 3D-printed keys, velocity sensing, and USB-MIDI/synthesizer modes.
- Engineered a **Hall-effect sensing system** using multiplexed analog inputs, reducing pin usage from 25 to 6.
- Developed Teensy firmware for sensor processing, MIDI handling, velocity detection, and real-time key control.
- Built a **self-playing system** using solenoids, MOSFET switching, and PWM for independent key actuation.

Quota — DeltaHacks12 First Place | *TypeScript, LangChain, MongoDB*

- Built a **TypeScript** VS Code extension for inline cost annotations, heatmaps, and one-click optimizations.
- Indexed **10,000+**-line codebases in under **3 seconds** using **AST**-based inefficiency detection.
- Cut developer API spend by up to **40%** by providing real-time API cost insights via regex and usage caching.
- Created a React Flow sandbox for budget-aware system planning with **RAG Gemini** chatbot.

Community

Hack the North Feb 2026 – Present
Logistics Organizer *Waterloo, ON*

- Led end-to-end operations for **300+ volunteers** supporting Hack the North, a **1,000+**-participant hackathon.
- Redesigned the volunteer shift-selection system to prevent assignment race conditions and streamline signup.
- Oversaw **300+** 3D-printed keychains from design to production given time and material constraints.